

Lab 32: Pandas DataFrame Creation and Combining

Aim

To create **Pandas DataFrames from dictionaries** and combine them using **concatenation and merging**.

Algorithm

1. Import the Pandas library.
2. Create two DataFrames using dictionaries.
3. Display the created DataFrames.
4. Combine the DataFrames using `concat()`.
5. Merge the DataFrames using a common column.
6. Display the combined results.

Program

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```
import pandas as pd
```

```
# Create first DataFrame
```

```
df1 = pd.DataFrame({  
    "ID": [1, 2, 3],  
    "Name": ["Alice", "Bob", "Charlie"]  
})
```

```
# Create second DataFrame
```

```
df2 = pd.DataFrame({  
    "ID": [1, 2, 3],  
    "Marks": [85, 90, 78]  
})
```

```

print("DataFrame 1:\n", df1)
print("\nDataFrame 2:\n", df2)

# Concatenate DataFrames
combined = pd.concat([df1, df2], axis=1)

print("\nConcatenated DataFrame:\n", combined)

# Merge DataFrames using ID
merged = pd.merge(df1, df2, on="ID")

print("\nMerged DataFrame:\n", merged)

```

Sample Output

DataFrame 1:

	ID	Name
0	1	Alice
1	2	Bob
2	3	Charlie

DataFrame 2:

	ID	Marks
0	1	85
1	2	90
2	3	78

Concatenated DataFrame:

	ID	Name	ID	Marks
0	1	Alice	1	85
1	2	Bob	2	90
2	3	Charlie	3	78

Merged DataFrame:

	ID	Name	Marks
0	1	Alice	85
1	2	Bob	90
2	3	Charlie	78

Result

Thus, Pandas DataFrames were successfully created from dictionaries and combined using concatenation and merging.